Carmen Galmes Altafulla

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Education

University of California, Santa Cruz, BS Physics (Astrophysics), BA in Studio Art — Sept 2022 – June 2026

- **GPA**: 3.79/4.0
- Relevant Coursework: Advanced Astronomy Lab, Intermediate Physics Lab, Physics of Planetary Systems, Physics of Stars, General Relativity, Thermodynamics and Statistical Mechanics, Electricity and Magnetism, Mechanics, Modern Physics, Quantum Mechanics, Introduction to Research in Physics, Introduction to Scientific Computing

Research Interests

Exoplanet Occurrence Rates and Evolution, Astrobiology, Transiting and Microlensed Exoplanets, Microlensing Star Populations

Research

TESS Exoplanet Demographics within a 100 Parsec Sample — March 2023 – Present
UCSC Astronomy & Astrophysics – Santa Cruz, CA

Advisors: Dr. Natalie Batalha, Dr. Anne Dattilo

- Used advanced statistical methods to analyze large datasets and identify patterns and trends in Jupyter Notebook
- Mathematically modeled the occurrence rates of exoplanets by joining and analyzing TESS and Gaia data products

Testing and Analysis of AC-LGAD Particle Sensors — Feb 2025 – Present
Santa Cruz Institute of Particle Physics – Santa Cruz, CA

Advisor: Dr. Simone Mazza

- Mounted, read, and analyzed different Low-Gain Avalanche Detectors (LGAD) using a LaserTCT and IV/CV system for further producing effective sensors for particle detection data-taking
- Analyzed RTCT data taken with Python code on a Linux OS

Identifying and Modeling Binaries with Microlensing Data — June 2025 – Present
Harvard & Smithsonian | Center For Astrophysics – Cambridge, MA

Advisor: Dr. Rosanne Di Stefano

- Worked on identifying binary star systems in microlensing data using lightcurve modeling and Chebyshev polynomials in anticipation of the Nancy Grace Roman telescope launch in early 2027
- Wrote a research report, presentation, and iPoster in tandem with research work

Experience

Large Group Tutor, UCSC Learning Support Services – Santa Cruz, CA — Sept 2023 – June 2024

- Facilitated discussions to help students develop communication and problem-solving skills
- Established positive relationships with students to create a safe and supportive learning environment
- Collaborated with professors and university administrators to effectively meet student needs

Resident Assistant, UCSC Res Life – Santa Cruz, CA — Sept 2024 – June 2025

- Attended, participated, and contributed to weekly staff meetings addressing resident needs
- Served as an approachable resource for residents seeking advice or assistance with personal or academic issues
- Responded to room transfers, incident reports, and maintenance requests
- Organized and hosted social and educational events for residents to foster meaningful connections and community spirit

Observing Experience

Advanced Astronomy Lab, 3 meter Shane Telescope at Lick Observatory - 1 night — Nov 2025

Publications

Characterization of the first full-size production for ePIC TOF layers — Nov 2025
Davis, M., Stage, G., Borgijin, A., ..., ***Galmes Altafulla, C.***, ..., et al.
arXiv:2511.02939

Volume-limited occurrence of TESS Hot Jupiters — In prep.
Galmes Altafulla, C., Jhamb, A., Dattilo, A., Batalha, N.

Presentations

Research Presentation: On the Missing Binary Source Population in Microlensing Events: Where are the Binaries? — Jun 2025
https://www.youtube.com/watch?v=tnQMFqj42RQ (Timestamp 4:13:00)

Research Poster: On the Missing Binary Source Population in Microlensing Events: Using Chebyshev Polynomials to Simulate Differences in Lightcurves — Jan 2026
247th American Astronomical Society Meeting in Phoenix, Arizona

Leadership and Broader Impacts

A Counter Space UCSC (President) — Sept 2024 – June 2026
- Held weekly meetings to share advice, ensure other members felt supported in their research, and within the department
- Arranged talks with weekly colloquium speakers and UCSC astronomy faculty on their experience as an underrepresented demographic in astrophysics
- Planned and held community events to support and foster community between underrepresented minorities in physics and astronomy

Artist for intro physics courses [PHYS 5A, PHYS 5B, PHYS 6B] — Sept 2025 – June 2026
- Collaborated with prof. Amy Furniss to supplement intro physics course education for more than 200 students
- Designed and created diagrams and representations of physical systems through artwork and animations

Peer Mentor- Intro to Research in Astronomy [ASTR 9A/9B] — Jan 2026 – June 2026
- Advised and mentored students on coding, presenting research, and research methods while being an approachable resource for general advice.
- Helped Dr. Jonathan Fourtney, and Graduate student Evan Davis lead project analyzing the boundary between large exoplanets, brown dwarfs, and dwarf stars.

Invited guest speaker at Harbor High school MESA club representing SCIPP and UCSC Astrophysics — Jan 2026

Invited Guest Speaker at Gardiner High School Career Day — Feb 2026

Awards

Osterbrock Rising Graduate Scholarship (2025), $500

Dean's List (5 quarters)

Eduardo Carrillo Memorial Scholarship (2026), $1000

NSF Graduate Research Fellowship Program (2026), Honorable Mention

Skills

Programming: Python, Astroquery, Pandas, Matplotlib, HTML, CSS, JavaScript

OS: Windows, Linux

Technologies: Microsoft Office (Word, Excel, PowerPoint, Teams), LaTeX, Adobe, Blender

Languages: English (fluent), Spanish (native), Catalan (proficient), French (proficient)